

Impact of Subway Surfer as a visual distraction on user during a reading comprehension task

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1 Motivation

Today there is a constant flow of visual distractions such as short-form video content, endless notifications and background games. Traditionally distractions are considered harmful and often lead to an decrease in productivity and focus. However, younger generations (Gen Z and beyond) have grown up with in an environment of constant multitasking, where studying while using your phone has become the new norm.

Has constant exposure to visual stimuli led to an adaptation in which younger people are less affected by distractions, or could these distractions even help them perform tasks better?

The objective of this experiment is to find out if the presence of a visual distraction, in this case a video of a Subway surfer constantly playing on part of the screen playing affects user performance and focus during the programming task. The experiment will be conducted in a controlled laboratory environment where participants complete a reading comprehension task. One group with subway surfer game play running simultaneously on part of the screen and the other without it. Participants will read short stories followed by multiple-choice questions to measure reading comprehension, while eye-tracking technology will be used to measure fixation durations and gaze patterns.

2 Experimental design

2.1 Goals, hypotheses and variables

The goal is to investigate the impact of a continuous visual distraction on the performance of a user during reading comprehension tasks.

Hypothesis:

H_0	H_a	Description (H_0)
$\rho(\text{DIS_T}, \text{ACC}) = 0$	$\rho(\text{DIS_T}, \text{ACC}) \neq 0$	There is no correlation between time spent looking at the distraction and on the comprehension accuracy
$\mu(\text{FIX_D_yes}) = \mu(\text{FIX_D_no})$	$\mu(\text{FIX_D_yes}) \neq \mu(\text{FIX_D_no})$	The mean fixation duration with distraction is equal to the mean without distraction
$\mu(\text{TASK_T_yes}) = \mu(\text{TASK_T_no})$	$\mu(\text{TASK_T_yes}) \neq \mu(\text{TASK_T_no})$	The mean time to complete the task is equal to the time without distraction
$\mu(\text{ACC_yes}) = \mu(\text{ACC_no})$	$\mu(\text{ACC_yes}) \neq \mu(\text{ACC_no})$	The mean of comprehensions accuracy with distraction is equal to the mean accuracy without distraction

Independent Variables:

	Name	Values	Description
Distraction	DISTR	Distraction, No Distraction	Whether a Subway Surfer video is playing on part of the screen or not

Dependent Variables:

	Name	Values	Description
Accuracy	ACC	Integer (%)	Percentage of correct answers on the comprehension questions after each reading task
Fixation Duration	FIX_D	Integer (ms)	Average duration of eye fixation during the task, tracked by eye-tracker.
Distraction time	DIS_T	Integer (seconds)	Total time participant was distracted by the distraction during a task.
Task Time	TASK_T	Integer (seconds)	Total time taken by the participant to complete the reading task

Controlled Variables:

Name	Values	Description
Age	18-21	All participants are first-year Computer Engineering students from the same university.
Language	Native	All participants have the same native language as the language of the tasks and school background language.

2.2 Design

During the experiment a participant completes the reading comprehension either with or without the visual distraction. Each participant will be given clear instructions on the task without informing them about the distraction, and the eye-tracking software will be calibrated.

2.3 Subjects

The subject of the experiment are 40 first-year engineer students from the same university ages 18-21 with the same tongue as the reading task, with a school background is also the same language. The students are ranked according to their results from their high school mother tongue exam results and then assigned to groups so that there is a even distribution of exam score in both groups.

2.4 Objects

- Subway surfer game footage on part of the screen (muted)

- Reading comprehension task
- Eye-tracking software and hardware
- timing and logging software

2.5 Instrumentation

- Eye-tracking software
- Screen recording
- interface for the task

2.6 Data collection

Each participant will complete one test either with the distraction or non-distraction condition. The duration of each test is approximately one hour, which is the length of a standard school lesson. Each test consists of 10 short texts, each followed by a multiple-choice question to assess reading comprehension. All data such as task completion time, eye-tracking metrics, and accuracy will be logged automatically by the software.

2.7 Analysis procedure

Analysis will be done according to the Jupyter notebook. The hypothesis with a mean will be tested using the t-test with a critical value for t of ± 0.139 to compare the means between the two groups, while for correlation the Pearson's correlation coefficient will be used with a critical value for r of ± 1.972 to measure relationship between accuracy and distraction time. If the p-value is below 0.05, we reject the null hypothesis.

2.8 Evaluation of validity

All participants are first-year engineering students at the same university, aged 18-21, with a shared native language, had the same school language before university, and the language of the reading tasks is in this native language for all participants. This is to minimize differences in language proficiency and educational background. Every participant gets the same exact task with the same exact questions, in the same exact sequence, so the difficulty of the task was the same. Every participant does the test in the same isolated room with the same exact setup, and given the same exact information about the task. Each participant is calibrated for the eye-tracking to ensure consistent data collection. The tests are done during weekday mornings to limit fatigue from the day, and the sessions are scheduled at the beginning of a new semester to minimize stress-levels and workloads affect on the study. While it is not possible to control a participant's sleep habits before taking part in the study each student is given an Oura ring to track their sleep the night before the day of the test and asked if they have slept well the last night, this is to ensure that energy levels of each student is as consistent as possible.